EdgeMatch Audit: Cross-Source Prerequisite Edge Corroboration as a Validation Signal

Executive Summary

EdgeMatch — the method of corroborating a prerequisite edge $a \to b$ from one educational corpus by finding a matching edge $c \to d$ in a second corpus, where $a \sim c$ and $b \sim d$ under a cross-source atom equivalence relation — is a theoretically coherent idea with clear precedent in educational data mining and knowledge graph alignment literature. However, the method carries strong structural assumptions that heterogeneous educational corpora routinely violate. A 2.7% strong direct-edge corroboration rate is **not a sign that the graph is mostly wrong**; it is the expected output of applying an intersection-heavy measure to corpora that differ in granularity, pedagogical scope, and instructional sequencing. The appropriate reading is that EdgeMatch, in its strict direct-edge form, is a **high-precision, low-recall corroboration signal**, not a coverage measure. Expanding it to transitive closure matching and building a null model baseline are the two most urgent next steps before drawing conclusions from the low rate.

1. Prior Work: Has EdgeMatch Been Done Before?

The closest analogue in the literature is the **concept-graph interlingua framework** of Liu, Yang, Ma, and Carbonell (2016). That JAIR paper explicitly maps courses from MIT, Caltech, Princeton, and CMU onto a shared concept vocabulary and then transfers prerequisite edges across institutions via the shared concept layer — which is structurally identical to EdgeMatch's endpoint-equivalence mechanism. The key finding was that concepts *are* partially shared across providers, but course-level prerequisites transfer imperfectly because the same concept pair may be encoded at different levels of granularity or scope in each institution's curriculum.[1][2]

Liang, Ye, Wu, Pursel, and Giles (2017) at Penn State proposed recovering concept prerequisites from course-level dependencies across 11 universities. Their CPR-Recover framework effectively treats cross-institution course dependency as a noisy signal for concept-level prerequisite truth — another form of multi-source corroboration. Domain-transfer experiments later showed that F1 drops severely when moving across domains, confirming that concept vocabularies and edge semantics do not port cleanly.[3][4] [^5]

The TutorialBank paper (Fabbri et al., 2018) manually annotated prerequisite chains across 200 NLP/ML topics drawn from multiple sources. It found **less than 40% topic overlap** between its manually selected topic list and a contemporaneous corpus (Gordon et al., 2016), even though both covered NLP. This directly mirrors your atom-equivalence problem: even within a single domain, independently constructed concept vocabularies share only a minority of nodes, so edge-level overlap is necessarily lower still.[^6]

The PREAP annotation protocol (Alzetta, Torre, and Koceva, 2023) documents that existing PR-annotated datasets "frequently report low annotation agreement and performance variability" and that inter-annotator kappa for prerequisite annotation is typically in the range of $\kappa = 0.30$–0.45 — categorized as only *fair* agreement. If human experts independently annotating the **same textbook** agree on only ~40% of edges under strict intersection, expecting a high direct-edge match rate across two independently extracted ML corpora is structurally unrealistic.[^7]

The LectureBank dataset (Li et al., 2019) collected 1,352 lecture files from 60 courses in NLP, ML, AI, deep learning, and IR across multiple domains and annotated 208 prerequisite concept pairs. Cross-domain prerequisite classification tasks built from LectureBank show substantial performance drops when training and test domains differ — providing quantitative evidence that the latent prerequisite structure of curricula does not uniformly transfer.[8][9] [^3]

PRELEARN (Alzetta et al., 2020), the first shared task on prerequisite learning, framed in-domain vs. cross-domain scenarios explicitly. The best system achieved an average F1 of 0.887 in-domain but only 0.690 cross-domain, a drop of nearly 20 F1 points — demonstrating that domain shift degrades the precision of prerequisite inference even with access to full textual content.[10][11]

2. Structural Assumptions EdgeMatch Requires

For direct EdgeMatch to produce a high corroboration rate, **all five of the following must hold simultaneously:**

2.1 Stable Concept Granularity

Both corpora must decompose the domain at the same level of abstraction. ARENA Chapter 0 is a practitioner-oriented engineering curriculum with hands-on coding atoms; Dive into Deep Learning is a textbook organized around mathematical derivations and algorithmic depth. A broad ARENA atom like "backpropagation" may split into "chain rule", "computational graph", "Jacobian-vector products", and "autodiff framework" in d2l. The atom equivalence map $\sim$ will fail to match these unless it explicitly handles supersets and subsets — but if it does, it introduces false positives in which unrelated concepts at the right embedding distance get merged.[12][7]

2.2 Comparable Instructional Scope

Two sources must cover the same *portion* of the deep learning curriculum. ARENA Chapter 0 is deliberately a foundations primer for ML-literate users approaching RL/interpretability; d2l covers foundational deep learning for a general ML audience. The intersection of their prerequisite graphs is limited to basic shared infrastructure (linear algebra, autograd, neural network fundamentals). Source-specific edges — covering reinforcement learning basics, transformer internals, or PyTorch idiom in ARENA, and covering custom training loops, modern architectures, or probabilistic models in d2l — will produce zero matches by construction, not by error.[2][1]

2.3 Consistent Edge Semantics

Prerequisite edges encode heterogeneous pedagogical relations. PREAP (2023) distinguishes between *absolutely necessary* prerequisites (strong PRs) and *useful-before* prerequisites (weak PRs). If one curriculum marks "linear algebra" as a formal prerequisite of "attention" and another marks it as a background recommendation, both sources encode a real prerequisite claim, but the edge semantics differ in strength and explicitness. EdgeMatch treats all directed edges identically, conflating these distinctions.[^7]

2.4 Reliable Atom Equivalence

The $\sim$ relation carries the entire load. The file `edge-matching-relation.md` explicitly acknowledges that lexical matching has a high false-negative rate, semantic cosine matching introduces a threshold hyperparameter, and LLM-judge definitions reintroduce the extraction bias you are trying to neutralize. The ontology matching literature shows that real-world KG alignment is hard even with rich relational context: even specialized models achieve only moderate alignment rates when the two graphs are structurally heterogeneous. Concept equivalence across curricula is a semantic alignment problem that no single simple measure solves reliably.[13][14] [^15]

2.5 Direct-Edge Representability

Both sources must encode the *same abstraction level* of the prerequisite chain. The document already identifies this blind spot: if source 1 teaches $a \to b \to c$ while source 2 teaches $a \to c$, neither edge matches the other, even though both express the same underlying pedagogical dependency at different resolutions. This is the canonical "abstraction alignment" failure mode documented in educational KG literature.[^16]

3. Why 2.7% Is Expected, Not Alarming

3.1 Intersection Geometry

Suppose ARENA has N_A atoms and d2l has N_D atoms. Let f be the fraction of atoms that are truly shared (i.e., truly equivalent concepts). The expected number of edge matches scales as f^2, because **both endpoints** must independently match. If only 30% of ARENA atoms have a genuine counterpart in d2l (plausible for a practitioner-vs-textbook pair), the theoretical ceiling on direct-edge match rate is $0.30^2 = 9\%$, and the actual rate will be lower due to noise in $\sim$.[1][6]

3.2 Empirical Precedent

The TutorialBank corpus found less than 40% topic overlap between two independently constructed ML/NLP concept vocabularies from the same research community, within the same decade. CPR-Recover evaluated on university courses from 11 institutions showed only modest recovery rates even when course-level prerequisites were used as direct supervision. CPRL's domain transfer experiments (Jia et al., 2021) show F1 drops from ~0.80 to ~0.39–0.43

when the model is applied across domains — meaning that even with a learned model, cross-domain prerequisite agreement is low.[4][6] [^3]

3.3 Human Expert Baselines Are Also Low

PREAP reports inter-annotator kappa of $\kappa \approx 0.30$ for prerequisite annotation with naive guidelines, improving to $\kappa \approx 0.60$ with PREAP's structured protocol — and this is for **the same textbook annotated by multiple annotators**. Cross-source annotation agreement, where annotators draw from different materials, would be lower still. Your 2.7% direct-match rate is the machine analogue of two experts annotating two different textbooks independently: the intersection of their edge sets will naturally be small.[^7]

3.4 Null Model

The appropriate null model is **random edge reassignment under a fixed degree sequence**. Specifically: given the actual atom equivalence map $\sim$, how many direct edges would match by chance if edges in d2l were randomly rewired? If the random-rewiring baseline also produces ~2–5% edge matches (which is likely given the small size of the equivalence map intersection), then 2.7% barely exceeds chance and EdgeMatch is nearly uninformative at this granularity. If 2.7% is substantially above the random baseline, it is already a meaningful signal — just weak in absolute terms.[17][18]

4. Failure Mode Taxonomy

The table below classifies the failure modes of direct EdgeMatch, their structural sources, and the appropriate remediation.

Failure Mode	Root Cause	Observable Signal	Remedy
Transitive bypass	Source 2 encodes $a \to b \to c$ while Source 1 has $a \to c$	Endpoints match but no direct edge	Transitive closure match
Concept splitting	Source 2 decomposes one ARENA atom into 3–5 sub-atoms	Source 2 has no atom equivalent to ARENA's broad atom	Hierarchical concept matching
Concept lumping	Source 1 aggregates what Source 2 treats separately	Spurious non-match despite semantic agreement	Partial/soft endpoint matching
Scope divergence	Genuinely source-specific topics	Atom is unmapped in cross-source equivalence	Source-Specific-Kept tier (correct behavior)
Ordering difference	Both cover A→B but different curricula order them differently	Edge exists in both but direction differs	Undirected matching as alternative
Edge-type mismatch	One source encodes "absolutely necessary" PR; other encodes "useful-before"	Weak semantic match without strong corroboration	Edge-type-aware matching
Equivalence noise	Atom equivalence over- or under-matches	False positives and false negatives in $\sim$	Ablations over similarity threshold τ

5. Literature-Supported Alternatives and Variants

5.1 Transitive Closure Matching (Highest Priority)

The EdgeMatch document already proposes this extension. It is the single most impactful upgrade to recall. PREAP explicitly notes that "non-annotated transitive relations remain implicit but can be inferred using PR properties" and recommends handling transitivity in agreement metrics. The expected effect: a substantial fraction of the unmatched ARENA edges that have both endpoints mapped will become matched because d2l encodes the same dependency through an intermediate atom. This upgrade tests the hypothesis that low recall is due to granularity mismatch rather than genuine disagreement.[^7]

5.2 Ancestor/Descendant Overlap

Rather than requiring a direct or transitive edge, check whether the equivalent d2l atoms have an ancestor/descendant reachability relation. This is a softer corroboration signal — it says "the same rough prerequisite ordering exists in d2l at some level of abstraction" — and will further increase recall while reducing precision. Useful as an upper bound on how much latent agreement exists.

5.3 Soft Graph Alignment via Embedding Similarity

Liu and Yang (2016) used a bipartite graph formulation to align concepts across institutions. Cross-lingual KG alignment methods (KGMA, 2019) use graph attention over topic entity graphs to match entities without requiring perfect 1:1 equivalence. Applying embedding-based entity alignment to the ARENA and d2l atom graphs would yield a probabilistic equivalence matrix rather than a hard binary $\sim$, enabling soft-score EdgeMatch. This is more expensive but removes the dependency on a threshold τ for atom equivalence.[^19][^1]

5.4 Hierarchical Concept Matching

The abstraction alignment literature (MIT CSAIL, 2025) proposes constructing an explicit "abstraction graph" that maps concepts at multiple levels of granularity and measuring alignment against it. For your setting, this means tagging atoms with an abstraction level (e.g., fundamental, methodological, implementation) and allowing matches across adjacent levels. An ARENA atom at level L would match a d2l atom at level $L\pm1$ if their definitions are semantically close. This directly addresses the splitting/lumping failure modes.[^16]

5.5 Multi-Source Voting with Three or More Corpora

With only two sources, a corroboration count of 0 or 1 is binary and noisy. Adding a third source (Karpathy's CS231n notes, Raschka's ML book, or a structured course like fast.ai) converts the corroboration count into a continuous variable and enables a majority-vote or rank-based tier. Liu and Yang (2016) explicitly advocate the multi-institution design: the concept interlingua improves in quality as more course graphs are aggregated. Three sources also enable a Krippendorff's α calculation, giving a defensible inter-source agreement estimate alongside the corroboration count.[^1]

5.6 Human-in-the-Loop Gold Sampling

The cleanest way to calibrate EdgeMatch is a targeted manual expert review. Sample ~50 edges that: (a) matched, (b) had both endpoints mapped but no matching edge, and (c) had one or zero endpoints mapped. Have an ML educator annotate each edge as "real prerequisite", "useful-before relationship", "questionable", or "invalid". This produces an empirical precision/recall estimate for the current EdgeMatch configuration and for the transitive closure extension, making both the absolute 2.7% rate and the tier boundaries interpretable.[20][7]

6. Evaluating the Proposed Tiering Scheme

The four-tier scheme proposed in the EdgeMatch document — Machine-Supported, Candidate, Source-Specific-Kept, Rejected — is methodologically defensible with three clarifications:

6.1 The Scheme Is Correct in Spirit

The two-axis design (intra-source robustness × cross-source corroboration) is structurally sound and mirrors the distinction between **reliability** (does the extraction replicate within one source?) and **validity** (does an independent source confirm the relationship?). This decoupling is methodologically stronger than using a single aggregate score, because an edge can be intra-source robust yet genuinely source-specific (as ARENA covering activation checkpointing optimization) without being "wrong."[21][12]

6.2 Upgrade the Corroboration Signal Before Finalizing Tiers

If you finalize the tier scheme using only direct-edge corroboration, you will systematically under-tier edges that fail due to transitive bypass or granularity mismatch — degrading them from Machine-Supported to Candidate when they should arguably be Machine-Supported. Run the transitive closure extension first and compare how many Candidate edges upgrade to Machine-Supported. If the number is large (>15–20% of all Candidates), the transitive version should be the primary corroboration signal; if it is small (<5%), the granularity-mismatch hypothesis is weak and the low corroboration rate likely reflects genuine domain divergence.

6.3 Normalize by Matched-Atom Coverage

The 2.7% rate is computed over all ARENA edges. It should be decomposed into three non-overlapping subsets:

1. **Both endpoints mapped** in the equivalence map (the only subset where direct or transitive EdgeMatch is even possible)
2. **One endpoint mapped** (partial coverage)
3. **Neither endpoint mapped** (completely source-specific)

The corroboration rate conditional on subset (1) is the informative quantity. If 2.7% of all edges match but 80% of ARENA atoms have no d2l equivalent, then 2.7% of all edges could equal ~14% conditional on coverage — a meaningfully different result that changes whether EdgeMatch is working correctly within its scope.[22][1]

6.4 Treat Source-Specific-Kept as a Positive Outcome

The current scheme uses "Source-Specific-Kept" as a fallback for unmatched edges. This is correct: many legitimate ARENA prerequisites (e.g., those around CUDA implementation, PyTorch internals, or RL environment interfaces) will never appear in d2l not because they are wrong but because d2l does not cover those topics. Treating source-specific edges with high intra-source robustness as legitimate is precisely what the literature recommends for heterogeneous curriculum settings.[6][1]

7. Priority Reading List

The following papers directly bear on whether cross-source edge corroboration is a valid validation signal:

Paper	Why Essential
Liu, Yang, Ma, Carbonell. "Learning Concept Graphs from Online Educational Data." *JAIR* 55 (2016).	Direct precedent: cross-institution concept graph interlingua; concept overlap rates across providers
Liang, Ye, Wu, Pursel, Giles. "Recovering Concept Prerequisite Relations from University Course Dependencies." *AAAI* 2017.	Recovery-based multi-source approach; domain coverage and sparsity data
Fabbri, Li, Tung, Radev. "TutorialBank: A Manually-Collected Corpus for Prerequisite Chains." *ACL* 2018.	Topic overlap <40% between independent ML/NLP vocabularies; directly models your atom-overlap problem
Jia, Shen, Tang, Sun, Lu. "Heterogeneous Graph Neural Networks for Concept Prerequisite Relation Learning." *NAACL* 2021.	Domain transfer failure of prerequisite models; confirms that cross-domain concept graphs diverge
Alzetta, Torre, Koceva. "Annotation Protocol for Textbook Enrichment with Prerequisite Knowledge Graph (PREAP)." *TKLL* 2023.	Baseline human inter-annotator agreement (~κ = 0.30–0.60); why low rates are normal for PR annotation
Roy, Madhyastha, Lawrence, Rajan. "PREREQ: Inferring Concept Prerequisite Relations from Online Educational Resources." *AAAI* 2019.	Best benchmark model; Siamese network on pairwise LDA representations
Alzetta et al. "PRELEARN: Prerequisite Relation Learning for Italian." *EVALITA* 2020.	Cross-domain drop from 0.887 to 0.690 F1; quantifies cost of domain shift in prerequisite extraction
MIT CSAIL. "Abstraction Alignment." 2025.	Formalism for comparing concepts across different abstraction levels; directly addresses granularity mismatch
Arora. "Incomplete Knowledge Graph Alignment." *arXiv* 2021.	KG alignment under structural heterogeneity and incompleteness; relevant to the $\sim$ definition problem

Paper	Why Essential
Pan, Li, Li, Tang. "Prerequisite Relation Learning for Concepts in MOOCs." *ACL* 2017.	Multi-feature approach; contextual + structural + semantic features for cross-course prerequisite inference

8. Concrete Recommendation

Keep EdgeMatch, but run it in two modes and interpret the two rates separately.

Mode 1 (Direct, as-is): High-precision, low-recall corroboration. An edge that survives direct EdgeMatch is a strong signal — it means two independently constructed curricula encode the exact same concept-level dependency at the same abstraction level. This is rare and therefore meaningful. The 2.7% rate is the precision-weighted corroboration floor.

Mode 2 (Transitive closure): Higher recall, lower precision corroboration. An edge that matches only under transitive closure is a weaker but still positive signal — the same dependency exists in the other curriculum but expressed through intermediate steps. Run this before finalizing tier assignments.

Decompose by coverage before reporting the rate. Compute the rate conditional on "both endpoints mapped." If this conditional rate is substantially higher (e.g., 10–20%), then EdgeMatch is working correctly within its scope and the apparent low overall rate reflects atom-level coverage limits, not edge-level disagreement.

Build the null model. Rewire d2l edges randomly (preserving degree sequence) and measure how many accidental matches occur under the current $\sim$ map. If the observed 2.7% is less than 2× the random baseline, EdgeMatch is not yet informative enough to justify tier elevation and you need to expand atom coverage or switch to transitive matching.

Add a third source. With a third ML curriculum (CS231n, fast.ai, Raschka's *Machine Learning with PyTorch and Scikit-Learn*), corroboration becomes a count variable (0/1/2) and multi-source intersection separates strong from weak corroboration rigorously. This is the most straightforward upgrade to the overall validation framework.

Your instinct is correct: EdgeMatch is most defensible as a **high-precision corroboration filter, not a recall measure**. The edges that survive it are almost certainly real; the edges that do not survive it are not thereby suspect — they are simply unconfirmed by this particular external source.

References

1. Learning Concept Graphs from Online Educational Data - The bi-directional mappings enable our system to perform interlingua-style transfer learning, e.g. t...

2. Concept Graph Learning : Networks Course blog for INFO 2040/CS ... - A novel machine learning technique called Concept Graph Learning, which can be used to predict prere...

3. [PDF] Heterogeneous Graph Neural Networks for Concept Prerequisite ...

4. Recovering concept prerequisite relations from university course ... - Liang C, Ye J, Wu Z, Pursel B, Giles CL. Recovering concept prerequisite relations from university c...
5. Recovering Concept Prerequisite Relations from University Course Dependencies - AAAI
6. TutorialBank: A Manually-Collected Corpus for Prerequisite Chains, Survey Extraction and Resource Recommendation - The field of Natural Language Processing (NLP) is growing rapidly, with new research published daily...
7. [PDF] Annotation Protocol for Textbook Enrichment with Prerequisite ... - The findings show that PREAP enables the creation of prerequisite knowledge graphs that have higher ...
8. LectureBank: a dataset for NLP Education and Prerequisite Chain Learning - Introduction In this blog post, we introduce our AAAI 2019 accepted paper “What Should I Learn First...
9. What Should I Learn First: Introducing LectureBank for NLP ... - Li, I., etc.(2019) introduced a new dataset, LectureBank, in 5 different domains with 208 manually-l...
10. NLP-CIC @ PRELEARN: Mastering Prerequisites Relations, from ... - We present our systems and findings for the prerequisite relation learning task (PRELEARN) at EVALIT...
11. PRELEARN @ EVALITA 2020: Overview of the Prerequisite Relation Learning Task for Italian - The Prerequisite Relation Learning (PRELEARN) task is the EVALITA 2020 shared task on concept prereq...
12. A systematic literature review of knowledge graph construction and ... - In the dynamic landscape of modern education, the search for improved pedagogical methods, enriched ...
13. Tailoring Knowledge Graph Embeddings for Ontology Alignment - Ontologies serve as structured, clearly defined representations of collectively agreed-upon concepts...
14. Survey on Embedding Methods Applied to Ontology Matching - Ontology matching plays a critical role in data integration, knowledge discovery, and ontology mergi...
15. [2112.09266] Incomplete Knowledge Graph Alignment - arXiv - In this work, we address the problem of aligning incomplete KGs with representation learning. Our KG...
16. Abstraction Alignment: Comparing Model-Learned and Human ... - Abstraction alignment externalizes domain-specific human knowledge as an abstraction graph, a set of...
17. [PDF] Parallel Generation of Simple Null Graph Models - A random graph is a graph with randomly wired edges, usually ... edge generation rate of about 1 bil...
18. [PDF] Null Models For Social Networks - Stanford Network Analysis Project - The Watts-Strogatz model [15] models all networks by connecting nodes to their nearest neighbors, th...
19. Cross-lingual Knowledge Graph Alignment via Graph Matching ... - In this paper, we introduce the topic entity graph, a local sub-graph of an entity, to represent ent...
20. ACE: AI-Assisted Construction of Educational Knowledge Graphs ... - Extraction of prerequisite relations among educational concepts from textual data is a challenging t...
21. ACE: AI-Assisted Construction of Educational Knowledge Graphs ... - We focus on a specialized type of Knowledge Graph called an Educational Knowledge Graph (EKG),

with ...

22. An Unsupervised Approach for Building the Concept Prerequisite ... - In this work, we propose an unsupervised approach, UPreG, for automatically inferring prerequisite r...